Lab #9: Plant Tissues

- Plants
  - Seed producing
    - angiosperms (flowring)
    - Gymnosperms (non-flowering)
  - not seed producing

* Flowering plants are devided into
  - monocotyledonous (monocot)
  - dicotyledonous (dicot)

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- Plant Tissue
  - Meristematic
  - Permanent
    - Simple
      - Parenchyma
      - Collenchyma
      - Sclerenchyma
      - Surface tissue (Epidermis)
    - complex
      - Xylem خشب
      - Phelom لحاء

Meristematic Tissues :-

- Small spherical or polyhedral to rectangular in shape, thin cell walls. ~~[illegible]~~

- Compactly arranged with no intercellular spaces.
- Each cell contains a dense cytoplasm and a prominent nucleus with small or no vacuoles.

** Meristematic tissues are classified as:-

1- Aptical Meristem :- growing tips of stems and roots, increasing the length of them, this meristem is responsible for the linear growth of an organ.

2- Lateral Meristem :- Causing the organ to increase in diameter and growth, lateral meristem occurs ~~the~~ beneath the bark of the tree to form Cork Cambium and in vascular bundles of dicots to form vascular cambium.

(*) The activity of this cambium results in the formation of secondary growth.

3- Intercalary Meristem :- This meristem is located between permanent tissues, ~~[illegible]~~ usually present at the base of node, inter node and on leaf base. Responsible for growth in length of plant and increasing the size of the internode. They result in branch formation and growth.

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(*) Permanent Tissues:

1- Simple permanent tissues:-

- Def: a group of cells that are similar in origin, structure, and function.

* Types of Simple Permanent Tissues:-

a- Parechyma:-

- Bulk of a substance with least specialized cells with thin layer of cytoplasm surrounding a central vacuole.
- Cell walls are usually loosely packed with intercellular spaces.

(*) Cells are alive.

- Chlorenchyma:- modified parenchyma containing chlorophyll and performs photosynthesis.

* Chlorenchyma found in palaside layer of leaf.

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b- Collenchyma:-

- is a living tissue of primary body like Parenchyma.
- Cells are thin-walled but posses thickening of cellulose, water and pectin substances.
- This tissue provides mechanical support, elasticity and tensile strength.
- It occurs chiefly in hypodermis of stems and leaves.
- It's absent in monocots and roots.

c- Sclerenchyma :-

- Contains of thick-walled , dead cells.
- These cells have hard and extremely thick secondary walls due to uniform distribution of lignin.
- closely packed without intercellular spaces between them.

- Found in hypodermis, pericycle, secondary xylem and phloem, endocarp of almond and coconut.

- The main function of this tissue is to provide hardness and protective covering to seeds and nuts.

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d- Epidermis (Surface Tissue):-

- The entire surface of the plant consists of a single layer of cells called epidermis or surface tissue.

- The outer and lateral walls of the cell are often thicker than the inner walls.

- The cells form a continuous sheet without inter cellular spaces.

- Protects all parts of the plant and it made the outer layer of stems, roots, and leaves.

2- Complex permanent tissues:-

- Contain of more than one type of cells work together as a unit.
- Help in the transportation of organic material, water and minerals up and down the plants.

- The common types of complex permanent tissue are:-

a- Xylem:-

- It's responsible for conduction of water and mineral ions / salts from the root to all parts of the plants.

- Composed of:- vessels, tracheids, fibers, parenchyma and ray cells.
- Tracheids have secondary cell wall tapered at the ends. They don't have end oppenings as vesseis and their ends overlap with pits that allow water to pass from cell to cell.

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b- Phloem:-

- Carries dissolved food substances throughout the plant.
- Composed of sieve-tube member and companion cells that are without secondary walls.
- The parent cells of the vascular cambium produce xylem & phleom.
- Sieve tubes are formed from sieve tube members laid end to end that don't have nuclei at maturity.
- The end walls are full of small pores called sieve plates where cytoplasm extends from cell to cell.
- The companion cells are nestled between sieve tube members that function in conduction of food.

Good Luck ^^

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